

Location, Carry and Policy Risk in the COMEX–London Gold Basis*

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Abstract

The difference between the COMEX gold futures price and the London benchmark is conventionally summarised as a single annualised rate. We show that this statistic is a horizon-weighted mixture of two economically distinct objects — a premium for metal being located in New York, which is a level, and a wedge between storage costs and the lease rate, which is a rate — and that the weight on the first, $365/T$, is set by the horizon at which the analyst chooses to read the futures curve. Estimating the curve daily across contracts over 2015–2026 separates the two, and three results follow. First, the measurement error induced by the three-and-a-half-hour gap between the London auction and the COMEX settlement is common across contracts and therefore loads entirely on the curve’s intercept: a regression of the estimated level on the day’s return yields 0.193 against the 0.146 implied by a random walk over the window, while the same regression on the slope yields -0.012 and is indistinguishable from zero. Second, the location premium averages $-\$0.55$ an ounce in calm years, inside a transfer-cost band independently estimated at $\$0.83$ – 10.25 , whereas the calm-period slope implies storage net of lease income of 0.58 percentage points a year — inconsistent with the 0.25% storage cost assumed throughout this literature. Third, the 2024–25 United States gold tariff episode was priced almost entirely as a level: the slope bounds the risk-neutral expected tariff at 0.92% even under a ten per cent arrival probability, and an over-identifying restriction linking slope and curvature to a tariff hazard is rejected.

Keywords: gold, commodity basis, covered interest parity, limits to arbitrage, trade policy, term structure.

JEL: G13, G14, F13, C22.

1 Introduction

Gold is traded simultaneously in two locations with different market conventions: in London as unallocated claims on metal held in vaults, and in New York as exchange-warranted metal deliverable against a futures contract. The difference between the two prices is the object that determines whether metal physically crosses the Atlantic, and it is therefore the price signal behind episodes in which very large gross bilateral flows appear in trade statistics without any corresponding change in ownership or consumption. The 2024–25 episode, in which roughly 535 tonnes moved from Switzerland to the United States and subsequently returned, is the motivating case.

That difference is almost always reported as an annualised rate, the “dislocation”: the futures-to-spot ratio converted to an implied financing rate and netted of a benchmark carry

*Replication: every number reported here is produced by `claude/spread-explainer/spread_model_checks.py` and the figure by `make_paper_figure.py`, both reading only files in `data/processed/`. The same system of equations, in the same numbering, appears in Section 9 of `SPREAD_EXPLAINED.html` and Section 4 of `spread_model_note.pdf`.

rate. This paper argues that the statistic is not a well-defined economic object. Writing the basis as the sum of a location premium, a policy-risk premium and a carry term makes the problem visible: annualising divides the first component by the horizon at which the curve is read while leaving the third unchanged, so the reported number mixes a level and a rate in proportions the analyst fixes by choosing a horizon. At thirty days the weight on the location premium is 12.2; at ninety it is 4.06; at one hundred and eighty it is 2.03. January 2025 reads 6.90, 2.83 and 1.84 percentage points at those three horizons respectively, from identical underlying data.

The alternative pursued here is to estimate the futures curve directly. On each date we fit the log settlement price across the traded contracts as a quadratic function of time to first notice, weighting by the square root of open interest. The intercept, net of the contemporaneous London price, estimates the location premium as a level. The slope, net of the dollar interest rate, estimates storage costs net of the lease rate. The curvature responds only to horizon-dependent policy risk. The decomposition requires no horizon convention, and its components are separately interpretable: the first is comparable to freight and refining costs, the second to quoted lease rates.

Four findings follow. First, the measurement error created by observing the two legs at different instants is a single common shift in every contract's log price on a given date, so it enters the intercept and not the slope (Proposition 1). This is testable and is borne out: the estimated level covaries with the day's gold return with a coefficient of 0.193 (standard error 0.015) against a random-walk benchmark of $3.5/24 = 0.146$, whereas the slope does not covary with it at all, -0.012 (standard error 0.011). Results resting on the slope therefore require no timing correction, and results resting on the level can be corrected by a reliability ratio of 0.80 at monthly frequency.

Second, in the calm years 2015–19 the estimated location premium averages $-\$0.55$ an ounce, or -0.047% of spot. Arbitrage thus holds the premium inside a narrow band around zero, as the theory requires, and the band's width is the transfer cost the threshold regressions in this literature attempt to recover; a threshold estimated from physical flows puts it at $\$3.77$ an ounce with a ninety per cent interval of $\$0.83$ – $\$10.25$, and trade sources put the all-in cost of moving London metal into COMEX-deliverable form at $\$1.5$ – $\$2.5$. The three readings are mutually consistent.

Third, the same calm period delivers a slope implying storage net of lease income of 0.58 percentage points a year. If the New York lease rate is near zero when nothing is happening, storage is of the order of 0.58%, not the 0.25% conventionally assumed; if storage is 0.25%, the implied lease rate is -0.33% , which is not a rate that is quoted. Because the assumed storage cost enters the annualised dislocation one-for-one, every reported *level* of that statistic inherits the error, and only differences are safe to interpret.

Fourth, the tariff episode was priced as a level on metal already inside the United States rather than as a term premium on future delivery. Inverting the slope for the risk-neutral expected tariff gives 0.92% under a ten per cent arrival probability within ninety days, 0.34% under twenty-five and 0.14% under fifty, against a tariff under public discussion at ten to thirty-nine per cent. The over-identifying restriction that links the slope and the curvature to a common hazard is rejected: it implies a near-certain tariff of one tenth of one per cent.

Related literature. The framework is the theory of storage (Kaldor, 1939; Working, 1949; Fama and French, 1987) applied to a commodity for which the convenience yield is unusually well identified, because above-ground inventories are enormous relative to flow demand and the own-rate of interest is directly quoted as a lease rate. Treating the futures price as a forward price abstracts from the margining wedge analysed by Cox, Ingersoll and Ross (1981). The location premium is the commodity analogue of a covered interest parity deviation, and the mechanism that sustains it — balance-sheet and financing costs borne by constrained

intermediaries — is the one emphasised in the currency literature by Du, Tepper and Verdelhan (2018) and Avdjiev, Du, Koch and Shin (2019), and more generally by Shleifer and Vishny (1997). The band structure in which arbitrage operates only beyond a transfer cost, and the resulting threshold in the flow response, is estimated in the manner of Hansen (2000). The relation between cash and futures prices as a price-discovery problem across venues follows Garbade and Silber (1983).

Section 2 describes the institutional setting and data. Section 3 sets out the model. Section 4 gives the estimating equations and the identification results. Section 5 reports estimates and tests. Section 6 discusses specification choices and limitations, and Section 7 concludes. Appendix A derives the estimating equations and proves the propositions.

2 Institutional setting and data

The London leg. The London Bullion Market Association Gold Price is set by auction, administered by ICE Benchmark Administration, twice daily; we use the afternoon auction, struck at 15:00 London time, in US dollars per troy ounce. The price refers to unallocated metal held in London vaults — a claim on a bullion bank’s pool rather than title to identified bars — and is retrieved from the LBMA’s public price service.

The New York leg. The COMEX gold futures contract calls for delivery of 100 troy ounces of metal in an exchange-approved vault in the New York area, conveyed as a warrant. Contracts are listed for a range of delivery months, of which February, April, June, August, October and December carry the great majority of open interest. For each contract and date we use the official settlement price, computed by the exchange from trading around 13:30 New York time, together with open interest. Both are taken from Databento’s `GLBX.MDP3` feed: settlement is statistic type 3 and open interest type 9. Settlement rather than closing price is the relevant quotation because it is the price at which positions are marked and against which an exchange-for-physical transaction is struck.

Rates and storage. The dollar interest rate is the Secured Overnight Financing Rate from 3 April 2018, spliced with the effective federal funds rate before that date, both from FRED. Storage is set to a constant 0.25% a year, an assumption rather than an observation: no free public series for vaulting and insurance exists. Section 5 shows the data reject this value.

Horizon and contract selection. Time to delivery is measured as τ , calendar days to first notice day, defined as the last business day of the month preceding the delivery month. First notice rather than last trade is the economically relevant horizon because from that date a long position may be assigned metal, so a financial holder must close or roll before it. A contract enters the daily cross-section if $15 \leq \tau \leq 400$ days, open interest is at least 1,000 lots, and a settlement price exists: inside fifteen days a contract is in its delivery window and no longer behaves as a forward, while far-dated or thinly held contracts carry stale settlements. This leaves a median of six contracts a day. Requiring both markets to be open loses 178 of 3,035 weekdays, predominantly around Christmas, New Year, Independence Day and Juneteenth. The estimation sample is 2,857 trading days, 2 January 2015 to 20 August 2026.

Table 1 collects the notation.

Table 1: Notation, and the status of each object

Symbol	Meaning	Status
S_t	London price, unallocated, 15:00 London	observed
$F_t(\tau)$	COMEX price for delivery in τ days	observed, 13:30 New York
$V_t(\tau)$	value of metal deliverable in New York at τ	latent
p_t	location premium, fraction of spot	latent; estimated
r_t	dollar interest rate	observed
s^N	New York storage cost, per year	assumed
ℓ_t^N, ℓ_t^L	lease rate on metal in New York, in London	latent
h_t, θ_t	tariff hazard rate; tariff rate if imposed	latent
$\Lambda_t(\tau)$	probability a delivery at τ is tariffed	latent
κ_t	all-in cost of moving an ounce between locations	latent; estimated
ε_t	gold's return from 15:00 London to settlement	latent; measurable
Q_t	metal shipped, tonnes a month	observed
$\bar{Q}$	refining and secure-freight capacity	latent

Notes. Rates are annualised decimals; differences between rates are reported in percentage points. Day count is 365 throughout. The superscripts N and L denote New York and London and are never powers. Prices are US dollars per troy ounce and flows are tonnes a month, where 1 tonne = 32,150.7 troy ounces.

3 The model

3.1 Carry within a location

A futures contract on metal held in New York is priced by the cost of carrying that metal to the delivery date. Writing $V_t(\tau)$ for the value today of metal deliverable in New York at horizon τ ,

$$F_t(\tau) = V_t(\tau) \cdot \exp \left[(r_t + s^N - \ell_t^N) \frac{\tau}{365} \right]. \quad (1)$$

The holder finances the position at r_t , pays storage s^N , and forgoes the lease income ℓ_t^N that metal in hand could earn if lent. Equation (1) is the gold analogue of covered interest parity, with the lease rate in the role of the foreign interest rate; it is the only equation in the system with a time dimension.

3.2 The location wedge and policy risk

Metal in New York need not be worth metal in London. Let p_t denote the premium on New York metal as a fraction of the London price, comprising location, bar form — London trades 400-ounce bars, COMEX takes 100-ounce and kilogramme bars — and the difference between an exchange warrant and a claim on a bank. Suppose in addition that an import tariff at rate θ_t may be imposed, arriving with constant hazard h_t , and that metal already inside the United States is exempt while metal imported for later delivery is not. Then

$$V_t(\tau) = S_t [1 + p_t + \Lambda_t(\tau) \theta_t] \quad (2)$$

$$\Lambda_t(\tau) = 1 - \exp \left(-h_t \frac{\tau}{365} \right) \approx h_t \frac{\tau}{365} - \frac{1}{2} \left(h_t \frac{\tau}{365} \right)^2, \quad (3)$$

where $\Lambda_t(\tau)$ is the risk-neutral probability that a delivery at τ is caught by the tariff. Equation (3) is the point at which the premium ceases to be a pure level: the expected tariff cost rises with the horizon, so policy risk enters the *slope* of the curve, and because Λ_t is concave in τ it also bends the curve downwards. The expansion requires $h_t \tau / 365 \ll 1$; if the hazard is large, Λ_t saturates and the policy term reverts to a level, a case Section 5 returns to.

3.3 The arbitrage band and the transfer cost

Arbitrage does not force p_t to zero. It confines it to a band whose edges are the cost of moving metal in each direction,

$$-\kappa_t^{N \rightarrow L} \leq p_t S_t \leq \kappa_t^{L \rightarrow N} \quad \text{unless capacity binds} \quad (4)$$

$$\kappa_t^{L \rightarrow N} = \underbrace{c_{\text{freight}} + c_{\text{ins}} + c_{\text{recast}}}_{\text{physical, per ounce}} + \underbrace{(r_t + \ell_t^L) \frac{d_{\text{transit}}}{365}}_{\text{financing in transit}} S_t. \quad (5)$$

The width of the band is therefore not a nuisance parameter but the quantity that the threshold in the flow equation identifies. Equation (5) is also the only point at which the *London* lease rate enters the system: shipping immobilises metal and money for the days in transit, so a London squeeze raises the cost of the arbitrage and widens the band, without shifting the COMEX curve. A single lease rate, as in (1), is therefore an approximation that fails precisely when the two markets are segmented.

3.4 Measurement

The two legs are not observed at the same instant. The London benchmark is struck at 15:00 London and the COMEX settlement approximately three and a half hours later, so the observed futures price embeds gold's return over the intervening window,

$$\widehat{F}_t(\tau) = F_t(\tau) \cdot e^{\varepsilon_t}, \quad \mathbb{E}[\varepsilon_t] = \mu, \quad \text{Var}[\varepsilon_t] = \sigma^2. \quad (6)$$

Because every contract settles at the same instant, ε_t is common across the cross-section on date t . This is the property that makes the measurement problem tractable, and Proposition 1 states its consequence.

3.5 Flows

Metal moves when the premium exceeds the cost of moving it, subject to the capacity of refineries and secure freight,

$$Q_t = \min\{\bar{Q}, \beta \max(0, p_t S_t - \kappa_t)\} + u_t. \quad (7)$$

The ceiling matters for interpretation: a premium far above the threshold can persist for months without constituting an arbitrage failure, because the technology for closing it is rate-limited.

4 Estimation and identification

Taking logarithms of (1) and (2), substituting (3) and (6) and collecting powers of τ (Appendix A) yields a curve that is quadratic in the horizon. The estimating equation, fitted by weighted least squares across the contracts available on each date with weights equal to the square root of open interest, is

$$\ln \widehat{F}_t(\tau_i) = \alpha_t + \beta_t \tau_i + \gamma_t \tau_i^2 + \eta_{it}, \quad (8)$$

and the three coefficients map onto the objects of Section 3 as

$$\hat{p}_t \equiv \alpha_t - \ln S_t = p_t + \varepsilon_t \quad (9)$$

$$\hat{\omega}_t \equiv 365\beta_t - r_t = s^N - \ell_t^N + h_t \theta_t \quad (10)$$

$$\gamma_t = -\frac{h_t^2 \theta_t}{2 \cdot 365^2} \leq 0. \quad (11)$$

The intercept is a level, measured in per cent of spot; the slope is a rate, measured in percentage points a year; the curvature is a bend, and is the only coefficient that responds to policy risk alone. Reassembling the components gives the spread at any horizon and, as a special case, the conventional statistic:

$$b_t(\tau) \equiv F_t(\tau) - S_t = S_t \left[\underbrace{p_t}_{\text{place, form, trust}} + \underbrace{\Lambda_t(\tau) \theta_t}_{\text{policy risk}} + \underbrace{(r_t + s^N - \ell_t^N) \frac{\tau}{365}}_{\text{carry}} \right] \quad (12)$$

$$\text{disloc}_t(T) \approx p_t \frac{365}{T} + h_t \theta_t + (s^N - s) - \ell_t^N. \quad (13)$$

Equation (13) is the paper's first point. Reading the fitted curve at a fixed horizon T and subtracting an assumed carry rate s divides the location premium by the horizon, leaves the policy and lease terms untouched, and adds the error in the storage assumption. The statistic is therefore a horizon-weighted mixture whose composition the analyst selects, not a parameter.

Proposition 1 (Measurement error loads on the level). *If ε_t is common across contracts on date t , then it enters the intercept of (8) one-for-one and does not enter the slope or the curvature. Consequently $\hat{p}_t$ is measured with error while $\hat{\omega}_t$ and γ_t are not, and $\text{plim cov}(\hat{\omega}_t, \varepsilon_t) = 0$.*

Proposition 2 (What the curve identifies). *From the daily cross-section alone: p_t is identified up to the additive measurement error ε_t ; the combination $s^N - \ell_t^N$ is identified whenever $h_t \theta_t = 0$; and the product $h_t \theta_t$ is identified relative to a period in which it is known to vanish. Neither s^N nor ℓ_t^N is separately identified, the decomposition of p_t into location, form and trust components is not identified, and ℓ_t^L does not enter the cross-section at all.*

Proposition 3 (Over-identification of the hazard). *Given a baseline value for $s^N - \ell_t^N$, equations (10) and (11) constitute two moments in the two unknowns h_t and θ_t . Writing $A \equiv h_t \theta_t$ and $B \equiv -2 \cdot 365^2 \gamma_t$,*

$$h_t = \frac{B}{A}, \quad \theta_t = \frac{A^2}{B},$$

so the model is over-identified: any third moment, or an independent estimate of either parameter, constitutes a test.

Proofs are in Appendix B.

5 Results

5.1 Scaling

Because the components are measured in different units, comparisons require explicit conversion, and Table 2 provides it. One percentage point of the conventional dislocation corresponds to 0.0822% of spot at a thirty-day horizon and 0.4932% at one hundred and eighty; at a gold price of \$2,756 those are \$2.27 and \$13.59 an ounce respectively. Conversely, a transfer cost of \$2 an ounce is 0.0726% of spot, equivalent to 0.29 percentage points when read at ninety days. A one-point movement in a ninety-day dislocation is thus worth roughly twice the entire cost of shipping the metal, which is the practical reason for preferring the level to the rate when the object of interest is a shipping decision.

Table 2: Converting between the units of the decomposition

	Per cent of spot	\$/oz at \$2,756	\$/oz at \$4,200
1 pp of $\text{disloc}_t(T)$, $T = 30$	0.0822	2.27	3.45
1 pp of $\text{disloc}_t(T)$, $T = 90$	0.2466	6.80	10.36
1 pp of $\text{disloc}_t(T)$, $T = 180$	0.4932	13.59	20.71
	Per cent of spot	pp read at 90 days	
\$1/oz of location premium		0.0363	0.15
\$2/oz, a representative transfer cost		0.0726	0.29
\$13.08/oz, the date in Table 3		0.4745	1.92

Table 3: The system evaluated on 29 January 2025

Equation	Quantity	Value
observed	S_t , London afternoon auction	\$2,756.30
observed	r_t	4.3500%
assumed	s , storage	0.25%
(8)	$\exp(\alpha_t)$, fitted curve at $\tau = 0$	\$2,769.38
(8)	$365\beta_t$, slope	5.2165%
(8)	γ_t , curvature	$-0.030 \times 10^{-6}/\text{day}^2$
(9)	$\hat{p}_t$, location premium	+0.473% of spot = +\$13.08/oz
(10)	$\hat{\omega}_t = s^N - \ell_t^N + h_t\theta_t$	+0.866 pp
(1),(2)	$F_t(90)$	\$2,805.23
(1),(2)	implied rate at ninety days	7.1991%
(12)	spread at ninety days	\$48.93/oz
	location component	\$13.08
	carry component	\$35.45
	compounding	\$0.40
(13)	$\text{disloc}_t(90)$	2.599 pp
	location, $\hat{p}_t \times 365/90$	1.919
	carry wedge, $\hat{\omega}_t - s$	+0.616
	cross-term	+0.063

5.2 The decomposition on a representative date

Table 3 runs 29 January 2025, near the peak of the tariff episode, through the whole system. The fitted curve at zero maturity stands at \$2,769.38 against a London price of \$2,756.30, so the location premium is \$13.08 an ounce or 0.473% of spot. The slope is 5.2165% a year against a dollar rate of 4.3500%, so storage net of lease income and any hazard term is 0.866 percentage points. Of the resulting \$48.93 spread at ninety days, \$35.45 is the time value of money and storage, \$13.08 is the location premium, and forty cents is compounding. The conventional statistic for that date, 2.599 percentage points, is three-quarters location premium — but only because ninety days was the horizon chosen.

5.3 The two components over time

Figure 1 plots monthly means of the two estimated components from 2019. The location premium is quiet for long stretches and moves sharply in three episodes: the spring of 2020, when freight and refining capacity collapsed; January 2025, during the tariff anticipation; and the late summer of 2025, following the customs ruling that reclassified bullion. The carry wedge moves on an entirely different rhythm, peaking in mid-2020 at roughly three percentage points and drifting through 2025 without a marked episode. That the two series

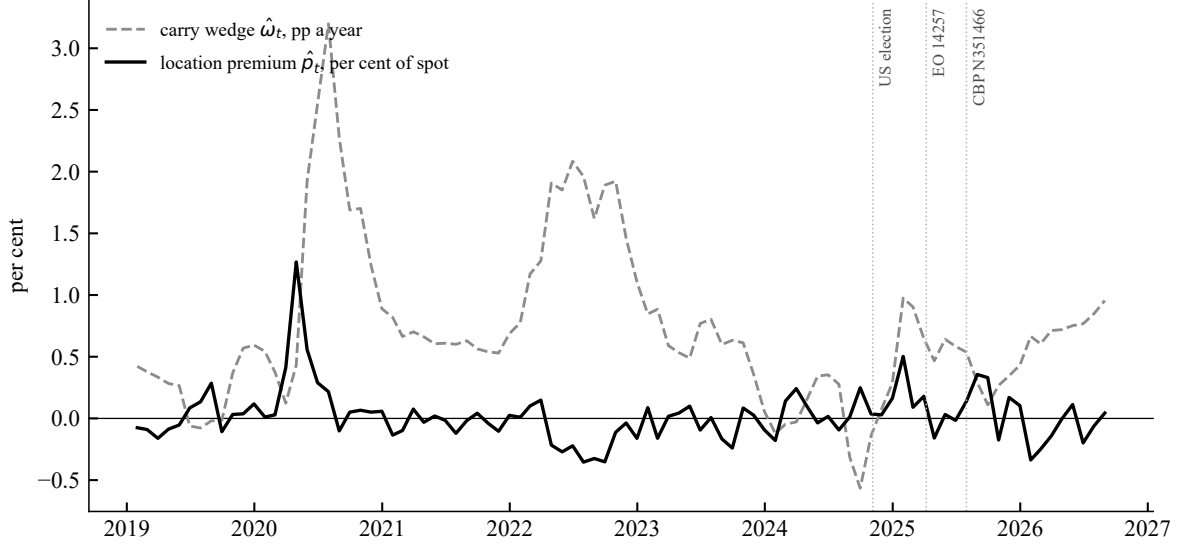


Figure 1: Monthly means of the location premium $\hat{p}_t$ and the carry wedge $\hat{\omega}_t$, 2019–2026

Notes. The location premium is the fitted curve’s intercept net of the contemporaneous London price, in per cent of spot; the carry wedge is the fitted slope net of the dollar interest rate, in percentage points a year. Monthly means of daily estimates; months with fewer than six trading days of data are left blank rather than interpolated. Vertical lines mark the 2024 United States election, the publication of Executive Order 14257, and US Customs and Border Protection ruling N351466.

Table 4: The components by period

Period	$\hat{p}_t$, % of spot	$\hat{p}_t$, \$/oz	$\hat{\omega}_t$, pp	Days
2015–2019 (calm)	−0.047	−0.55	+0.58	1,230
2019 alone	+0.011	+0.21	+0.25	247
Nov 2024–Apr 2025 (tariff)	+0.137	+3.82	+0.56	120
January 2025 alone	+0.502	—	+0.98	21
Aug–Oct 2025 (customs ruling)	+0.157	+5.22	+0.22	64
2026	−0.111	−5.21	+0.75	157

are only loosely related is the empirical content of the decomposition: a single annualised statistic necessarily blends them.

Table 4 reports period means. Across 2015–19 the premium averages −0.047% of spot, or −\$0.55 an ounce: economically zero, and inside the transfer band of (4). During the tariff anticipation window it averages +0.137% (\$3.82) and in January 2025 alone +0.502%. Following the customs ruling it averages +0.157% (\$5.22), while in 2026 it turns negative, −0.111% (−\$5.21), New York trading cheap as metal returned. The carry wedge, by contrast, is positive throughout and highest in 2026, when the premium is at its most negative.

5.4 Testing the measurement model

Proposition 1 has a direct empirical implication, reported in Table 5. Regressing the estimated level on the contemporaneous London-to-London return yields a coefficient of 0.193, close to but somewhat above the 0.146 that a random walk over three and a half of twenty-four hours implies; the excess is what one expects if the United States morning carries more than a time-proportional share of the day’s variance. The same regression for the slope yields −0.012 with a standard error of 0.011, which is the prediction of the proposition. A placebo using the previous day’s return returns 0.049 (standard error 0.017), small but not zero, indicating mild persistence that a purely classical error model does not capture.

Table 5: The timing error loads on the level, not the slope

Dependent variable	Same-day return	Previous-day return	R^2
$\hat{p}_t$, location premium	0.193 (0.015)		0.155
$\hat{p}_t$, location premium	0.194 (0.015)	0.049 (0.017)	
$\hat{\omega}_t$, carry wedge	-0.012 (0.011)		0.000
Random-walk benchmark for the same-day coefficient: $3.5/24 = 0.146$			

Notes. Daily observations, 2015–2026, $n = 2,856$. Heteroskedasticity-robust standard errors in parentheses. The return is the log change in the London afternoon auction price from date t to $t + 1$, which contains the 15:00-to-settlement window of date t .

Scaling up, the daily standard deviation of the London benchmark is 1.01%, implying 0.39% of spot, or \$7.44 an ounce, of measurement noise in a single day’s level; the measured standard deviation of the level is 0.50%, so the timing gap accounts for roughly sixty per cent of its daily variance and forty-seven per cent in 2019 alone. At monthly frequency, averaging twenty sessions reduces the noise to 0.086% against a monthly series standard deviation of 0.192%, a reliability ratio of 0.80: slope coefficients estimated on the monthly level are attenuated by a fifth and should be scaled by 1.25. The bias is bounded below by \$0.31 an ounce under a uniform-drift assumption, which is negligible relative to an episode premium of \$14 but not relative to a calm mean of -\$0.55. Re-timing the New York leg using intraday prices at the auction instant would eliminate the problem rather than correcting for it.

5.5 Storage and the lease rate

The calm-period slope implies $s^N - \ell_t^N = +0.58$ percentage points a year. The two cannot be separated without external information (Proposition 2), but neither admissible resolution is comfortable. If the New York lease rate is approximately zero when the market is quiet, storage is of the order of 0.58% a year rather than the 0.25% assumed; if storage is indeed 0.25%, the lease rate implied is -0.33%, which is not a rate that is quoted in this market. Because the assumed storage cost enters (13) one-for-one, the level of every reported dislocation is displaced by the error. Differences are unaffected, and until an external vaulting and insurance schedule is obtained, only differences should be interpreted.

5.6 Was the tariff episode priced in the term structure?

Proposition 3 allows the hazard and the tariff rate to be recovered separately, given a baseline for $s^N - \ell_t^N$. Using 2015–19 as the baseline ($\hat{\omega} = 0.58\%$, $\gamma = -0.007 \times 10^{-6}$), January 2025 delivers $A = h_t \theta_t = 0.00395$ a year. Combining this with the observed curvature yields h_t between 3.4 and 3.9 a year — a fifty-seven to sixty-two per cent probability of imposition within ninety days — paired with θ_t between 0.10 and 0.12 per cent. A near-certain tariff of one tenth of one per cent does not describe the policy under discussion, and the over-identifying restriction is therefore rejected. The natural reading is that the curvature is dominated by ordinary term-structure variation rather than by a policy hazard: it is already negative in the calm years, where by construction no tariff risk exists.

Imposing a hazard instead and inverting (10) for the tariff rate bounds the risk-neutral expectation, and the bound is small: a ten per cent probability within ninety days implies $\theta_t = 0.92\%$, twenty-five per cent implies 0.34%, and fifty per cent implies 0.14%. Over the full anticipation window, November 2024 to April 2025, $\hat{\omega}$ averages +0.561 percentage points, *below* the calm baseline, so the slope prices no positive hazard at all. Against a tariff under

public discussion at ten to thirty-nine per cent, the term structure was pricing almost none of it.

The conclusion is not that the episode was imaginary but that it was priced as a level on metal already inside the United States rather than as a term premium on future delivery. This is consistent with the accounting in Table 4: measured against the five-year calm mean rather than a single quiet day, the level accounts for approximately eighty-five per cent of the January 2025 movement in the conventional statistic.

5.7 Flows

The threshold in (7), estimated on monthly Swiss customs shipments to the United States over 2023–26, is \$3.77 an ounce with a ninety per cent interval of \$0.83–10.25; below it, flows sit on a floor of two to three tonnes a month, and in the highest premium bucket the median is sixty-four tonnes. The point estimate is consistent both with the calm-period location premium and with trade estimates of \$1.5–2.5 an ounce for freight, insurance and recasting, which is the joint validation the band in (4) makes available. The coefficient itself, however, is not structural: shipments respond to the premium and simultaneously close it, so a monthly regression recovers an equilibrium relation rather than the supply of arbitrage.

6 Specification and limitations

The reading horizon. Although the decomposition requires no horizon, the conventional statistic does, and the choice is not innocuous. Three criteria are measurable. First, support: the shortest contract in the daily cross-section is a median of thirty days out and at most seventy-seven, so a horizon of seventy-seven days or more lies inside the observed data on every date in the sample, whereas a thirty-day reading extrapolates below the shortest available contract on 48.3% of dates. Second, precision: the weighted centre of mass of the cross-section averages eighty-two days, and on a grid from twenty to two hundred and seventy days the variance of the fitted value is minimised at ninety. Third, specification: allowing the quadratic term moves the fitted value least, in dollar terms, near ninety days (\$0.33 an ounce, against \$0.46 at thirty and \$0.40 at one hundred and eighty). Ninety days is also the tenor at which the covered interest parity literature conventionally quotes deviations. These criteria justify the convention without rescuing the statistic.

Weighting. Square-root-of-open-interest weights are ad hoc; the defensible weight is the inverse variance of the settlement error, for which open interest is a proxy of unknown functional form. In practice the most heavily weighted contract carries a median 46% of the total weight, with a middle half of 43–51%, so the fitted curve is less independent of the front contract than the term “constant maturity” suggests.

Futures against forwards. Equation (1) treats the futures price as a forward price, abstracting from the margining wedge of Cox, Ingersoll and Ross (1981). The wedge is small at these horizons but falls on β_t , which is where the lease rate is read.

Aggregation within a location. The model admits one New York price, so the registered and eligible warrant categories, and the difference between 100-ounce and kilogramme bars, are absorbed into p_t . A single lease rate per location, flat in τ , is likewise an approximation; a sloped lease curve would appear as curvature and contaminate Section 5.

Functional form. A third-order term in (8) would reallocate weight between β_t and γ_t , which is precisely the margin on which Proposition 3 is tested. The rejection reported above should be read as a rejection of the joint hypothesis that the curve is log-quadratic and that its curvature is generated by a tariff hazard.

The flow measure. Swiss customs shipments omit direct London-to-New York movements of already-deliverable bars, and non-Swiss origins. If the omitted flows respond to the same signal, β is attenuated and the estimated threshold biased.

7 Conclusion

The conventional annualised gold basis is a mixture of a level and a rate whose composition is determined by an analytical choice rather than by the market. Estimating the futures curve directly separates the two at no cost in data: the intercept identifies a location premium comparable to physical transfer costs, and the slope identifies storage net of the lease rate, comparable to quoted lease rates. Doing so shows that the measurement error induced by non-synchronous observation is confined to the level, that the storage cost assumed in this literature is inconsistent with the term structure in calm periods, and that the 2024–25 tariff episode was a level phenomenon that the term structure barely priced.

Three extensions would sharpen the analysis at low cost. Re-timing the New York leg with intraday prices at the auction instant would remove rather than attenuate the measurement error. An external vaulting and insurance schedule would break the storage-lease identification problem and thereby fix the level of every dislocation series in this literature. Finally, the dated resolutions of the policy episode — the April 2025 exemption and the July 2025 customs ruling — provide events at which the hazard falls to zero by construction, and hence a sharper test of whether policy risk ever entered the slope than the period comparisons used here.

A Derivation of the estimating equation

Substituting (2) into (1), multiplying by the measurement factor in (6) and taking logarithms,

$$\ln \hat{F}_t(\tau) = \ln S_t + \ln[1 + p_t + \Lambda_t(\tau)\theta_t] + (r_t + s^N - \ell_t^N) \frac{\tau}{365} + \varepsilon_t. \quad (\text{A.1})$$

For small p_t and $\Lambda_t\theta_t$ the logarithm is approximately its argument less one, and substituting the second-order expansion of Λ_t from (3) gives

$$\ln \hat{F}_t(\tau) \approx \underbrace{\ln S_t + p_t + \varepsilon_t}_{\alpha_t} + \underbrace{\left[\frac{r_t + s^N - \ell_t^N}{365} + \frac{h_t\theta_t}{365} \right]}_{\beta_t} \tau - \underbrace{\frac{h_t^2\theta_t}{2 \cdot 365^2}}_{-\gamma_t} \tau^2, \quad (\text{A.2})$$

which is (8) with the coefficient mappings (9), (10) and (11). Two features are worth noting. The location premium and the measurement error are indistinguishable within a date because both are constants in τ ; and the policy term contributes to both the linear and quadratic coefficients, which is what makes Proposition 3 available.

B Proofs

Proof of Proposition 1. By (6) the observed log price on date t is $\ln F_t(\tau_i) + \varepsilon_t$ for every i in the cross-section, so ε_t is collinear with the constant regressor in (8) and orthogonal to τ_i and τ_i^2 within the date. Weighted least squares therefore attributes it entirely to α_t , leaving

β_t and γ_t unchanged. Since $\hat{\omega}_t$ is a function of β_t and the observed r_t alone, it is uncorrelated with ε_t by construction. $\square$

Proof of Proposition 2. Immediate from (9), (10) and (11). Equation (9) gives $p_t + \varepsilon_t$, so p_t is identified only up to the error, and consistently under averaging if $\mathbb{E}[\varepsilon_t] = 0$. Equation (10) gives the sum $s^N - \ell_t^N + h_t\theta_t$; with $h_t\theta_t = 0$ it gives $s^N - \ell_t^N$, but no function of the cross-section separates s^N from ℓ_t^N , since both enter only through that sum. Given a period in which $h_t\theta_t = 0$, the difference of (10) between periods identifies $h_t\theta_t$ under the assumption that $s^N - \ell_t^N$ is unchanged. The three components of p_t enter (2) only through their sum. The London lease rate ℓ_t^L appears in (5) alone, which involves no quantity in the New York cross-section. $\square$

Proof of Proposition 3. Write $A = h_t\theta_t$ from (10) net of the baseline and $B = -2 \cdot 365^2 \gamma_t = h_t^2\theta_t$ from (11). Then $B/A = h_t$ and $A/h_t = A^2/B = \theta_t$, provided $A \neq 0$ and $B > 0$. Both moments derive from the same fitted curve, so any additional restriction — a third moment, an independent estimate of h_t or θ_t , or a sign restriction such as θ_t lying in a plausible range — is testable. $\square$

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